particles and the familiar world of people and their equipment, the rules change because the sizes change. The motivation for asserting this division is clear: a tiny particle, according to quantum mechanics, can be located in a fuzzy mixture of here and there, yet we don't see such behavior in the big, everyday world. But exactly where is the border? And, of vital importance, how do the two sets of rules interface when the big world of the everyday confronts the minuscule world of the atomic, as in the case of a measurement? Bohr forcefully declared these questions to be out of bounds, by which he meant, truth be told, that they were beyond the bounds of what he or anyone else could answer. And since even without addressing them the theory makes astonishingly accurate predictions, for a long time such issues were far down on the list of critical questions that physicists were driven to settle.

But to understand quantum mechanics completely, to determine fully what it says about reality, and to establish what role it might play in setting a direction to time's arrow, we must come to grips with the quantum measurement problem.

In the next two sections, we'll describe some of the most prominent and promising attempts to do so. The upshot, should you at any point want to rush ahead to the last section focusing on quantum mechanics and the arrow of time, is that much ingenious work on the quantum measurement problem has yielded significant progress, but a broadly accepted solution still seems just beyond our reach. Many view this as the single most important gap in our formulation of quantum law.

Reality and the Quantum Measurement Problem

Over the years, there have been many proposals for solving the quantum measurement problem. Ironically, although they entail differing conceptions of reality—some drastically different—when it comes to predictions for what a researcher will measure in most every experiment, they all agree and each one works like a charm. Each proposal puts on the same show, even though, were you to peek backstage, you'd see that their modi operandi differ substantially.